

Avnish Raj

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SUMMARY

Software Engineer and B.Tech Computer Science student building full-stack applications, AI systems, developer tools, and secure web platforms. Experienced with TypeScript, Python, React, Next.js, Node.js, REST APIs, cloud databases, authentication, automated testing, and production-oriented application design.

EDUCATION

Thapar Institute of Engineering and Technology

B.Tech in Computer Science; CGPA: 7.68/10

Patiala, India

2024 – 2028

TECHNICAL SKILLS

Languages: C++, Python, JavaScript, TypeScript, SQL, Solidity

Frontend: React, Next.js, Vite, Tailwind CSS, Progressive Web Apps, Responsive UI

Backend: Node.js, Express.js, REST APIs, API Routes, Authentication, Authorization

Databases and Cloud: MySQL, Firebase, Cloud Firestore, Supabase, Cloudinary, Vercel, Base Sepolia

AI and Tools: AI/LLM Integration, Groq, Hindsight Cloud, Zod, Vitest, Git, GitHub, JSONL, Unix PTY

Core CS: Data Structures and Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Security

PROJECTS

Lifeline | Python, CLI, Unix PTY, JSONL, Security

[GitHub](#)

- Developed a dependency-free Python CLI that captures interrupted AI coding sessions and resumes work across Claude Code, Codex, and Gemini without manual context reconstruction.
- Engineered cross-tool session readers, automatic usage-limit detection, JSONL persistence, restrictive file permissions, secret redaction, path validation, and argv-based process launching.

SCAR: Production's Immune System | Next.js, TypeScript, React, Groq, Zod, Vitest

[GitHub](#) | [Live](#)

- Built a deployment-risk AI agent that stores production incidents, retrieves causally similar failures, and improves future deployment verdicts through persistent organizational memory.
- Implemented server-side API routes, structured AI risk analysis, evidence-citation gates, isolated demo sessions, deterministic fallback logic, secure request handling, and 17 automated tests.

Campus Lost and Found PWA | JavaScript, Firebase Auth, Firestore, Cloudinary, PWA

[GitHub](#) | [Live](#)

- Created and deployed a campus-exclusive Progressive Web App for reporting and recovering lost items with intelligent matching, real-time notifications, search, filtering, and moderation.
- Integrated restricted Google authentication, Cloud Firestore persistence, Cloudinary image uploads, client-side compression, auto-expiring posts, and an installable service-worker experience.

UGF Skill Passport | React, TypeScript, Solidity, ethers.js, Base Sepolia, Vite

[GitHub](#) | [Contract](#)

- Built a Web3 quest application that mints onchain NFT skill badges through a deployed Solidity smart contract with verifiable onchain proof.
- Integrated wallet connection, Base Sepolia network handling, UGF gas abstraction, alternate-token fee settlement, duplicate-claim prevention, and transaction feedback flows.

RELEVANT COURSEWORK

Data Structures and Algorithms, Database Management Systems, Object-Oriented Programming, Operating Systems, Software Engineering, Secure Coding, API Development